Mohammad Fardeen Shaik

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AI SOFTWARE ENGINEER

Education

George Mason University, College of Engineering **Aug 2025 – May 2027**
M.S. Computer Science
Coursework: Intro Artificial Intelligence,Software Eng for WWW, Component based Software Devolepment,Analysis of Algorithms,Computer Systems and System Programming,Mathematical Foundations of CS,Database Ssytems,Mobile Immersive Computing,Modern Computer Architecture

Vellore Institute of Technology **Sept 2021 – May 2025**
B.Tech. Computer Science and Engineering
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, AI & ML

Technical Skills

Languages: Python, JavaScript, Java, SQL, R, HTML, CSS
Tools: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebooks
Core Areas: Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, Statistics, Data Analysis, Feature Engineering

Professional Experience

Quadrant Technologies | Cloud AI Intern | Redmond,WA Jun 2026 – Aug 2026

- Developed an explainable candidate ranking system using semantic search and LLMs with adjustable scoring weights across resume; implemented blind review mode showing exactly where each matched skill, ensuring full transparency trust in AI decisions.
- Built Screening Assistant using React, FastAPI, and Azure services to automate resume parsing; leveraged Azure AI Document Intelligence and Azure OpenAI for intelligent extraction enabling recruiters to reduce manual screening time significantly.

Ethnus Codemithra | Full-Stack Developer Intern Aug 2023 – Nov 2023

- Enhanced UI/UX by refactoring reusable React components and optimizing state management, resulting in 15% increased user engagement; implemented automated tests with Jest/React Testing Library, API monitoring with Postman Collections, and achieved 80

Projects

AI Contract Invoice Intelligence System |Apache Airflow, FastAPI APIs, PDF Processing Pipeline, Database Integration Nov 2025

- Built scalable NLP pipelines using Python and TensorFlow to automate invoice extraction and data enrichment from unstructured PDFs; successfully processed 150,000+ invoices annually with 98% accuracy, verified through quarterly QA audits.
- Engineered Apache Airflow DAG-based orchestration workflow covering end-to-end ingestion, text extraction, validation, and storage pipelines; implemented automated validation rules and error detection that reduced data quality issues by 60%.

Blood Bank Management System | React, FastAPI, MySQL, Google Cloud Platform (GCP), RESTful APIs Oct 2025

- Developed full-stack cloud-based donor registration and blood inventory management system using React, FastAPI, and MySQL, deployed on Google Cloud Platform with real-time availability search and request tracking.
- Engineered RESTful APIs for donor onboarding, blood donation history, and inventory management; implemented Google Cloud Storage integration for document handling and optimized database queries reducing query latency by 40%.

Loan Default Prediction | Streamlit, Python Backend, SHAP/LIME Explainability, Data Visualization,PyTorch, XGBoost Feb 2026

- Developed an ensemble machine learning model combining PyTorch neural networks and XGBoost gradient boosting to predict loan defaults; performed extensive feature engineering, hyperparameter tuning, and cross-validation to achieve high predictive accuracy.
- Deployed an interactive Streamlit web application powered by SHAP and LIME explainability frameworks, enabling non-technical banking professionals to understand model predictions and make transparent, data-driven lending decisions with full audit trails.

Autism Detection | Flask Web Framework, CNN/LSTM Models, MySQL Database, Frontend UI,Python, TensorFlow Apr 2026

- Designed and trained CNN/LSTM hybrid deep learning architecture for autism spectrum disorder screening; enhanced baseline model accuracy by 15% to achieve 91% using advanced training techniques learning-rate scheduling, and class-weight balancing.
- Built end-to-end deep learning pipeline with data preprocessing, model training, validation, and evaluation on 1,500+ labeled medical samples; integrated Flask backend with MySQL database for patient data management and integrated model inference.

Research

Hybrid Algorithm for Parkinson's Prediction | Python, Scikit-Learn (Random Forest, SVM), SQL. Nov 2025

- Presented hybrid machine learning model at Springer 2024 combining Random Forest and SVM algorithms, achieving 88.7% accuracy on UCI Parkinson's dataset (195 instances) with 6-8% improvement over individual models by optimizing sensitivity and specificity for early-stage neurological disorder detection.

Leadership

IEEE Computer Society — Events Head (Mar 2022 – Dec 2024): Led large-scale technical events and hackathons with participation exceeding 700 students.